Physics 468/668: Statistical Physics **Spring 2019**

Lecture 2: The Statistical Basis of Thermodynamics

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2.1 Microstates and Macrostates

We consider a system composed of N identical particles confined to a space of V. The total energy E would be equal to the sum of the energies of the individual particles.

$$E = \sum_i n_i \epsilon_i \tag{2.1}$$

The specification of the actual values of the parameters N, V, E then defines a *macrostate* of the system. At the molecular level, however, a large number of possibilities still exist because at that level there will be a large number of different ways to make the total state of N, V, E (think out arranging the coins with different sequence of head and tail). Each of the different ways specifies a *microstate* or *complexion* of the given system.

The actual number of all possible microstates (Ω) will be a function of N, V, E. In principle, it is from the magnitude of the number of Ω and from its dependence on the parameters N, V, E, that complete thermodynamics can be derived.

2.2 Multiplicity in Einstein Solids

N: Number of the oscillators.
q: Number of energy states.

$$\Omega(N, q) = \binom{q + N - 1}{q} = \frac{(q + N - 1)!}{q!(N - 1)!} \tag{2.2}$$

It can be simply proved as follows
q circles;
$N - 1$ vertical lines;
how to arrange them?



Exercises
Calculate the multiplicity of an Einstein solid with 5 oscillators and [1,2,3,4,5] units of Energy.

q	$\Omega(5,q)$
1	
2	
3	
4	
5	

2.3 Contact between statistics and thermodynamics

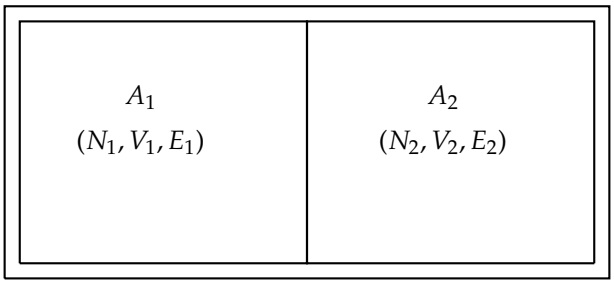


Figure 2.1: A schematic of two physical systems in thermal contact.

Let's first figure out how Ω is related to the thermodynamic quantities. We consider two physical systems, A_1 and A_2, which are separately in equilibrium. Let the macrostate of A_1 be represented by the parameters N_1, $V_1$and E_1 so that it has $\Omega_1(N_1, V_1, E_1)$ possible microstates, and the macrostate of A_2 be represented by $\Omega_2(N_2, V_2, E_2)$. Can we derive some thermodynamic properties from $\Omega_1(N_1, V_1, E_1)$ and $\Omega_2(N_2, V_2, E_2)$?

Let's bring two systems into thermal contact. For simplicity, we only allow the heat exchange between the two, while N, V remain fixed. This means there could be some interchanges between E_1 and E_2, however, it has to be restricted by the conservation law.

$$E = E_1 + E_2 = \text{const} \tag{2.3}$$

From the microscopic view, the total number of microstates could be expressed as,

$$\Omega_1(E_1)\Omega_2(E_2) = \Omega_1(E_1)\Omega_2(E - E_1) \tag{2.4}$$

When the system approaches to the equilibrium, what should be the value of $\bar{E_1}$? According to the 2nd law, the entropy should reach the maximum. Mathematically, we need to find $\bar{E_1}$ which satisfies,

$$\left(\frac{\partial\Omega_1(E_1)}{\partial E_1}\right)_{E_1=\bar{E_1}}\Omega_2(E_2) + \left(\frac{\partial\Omega_2(E_2)}{\partial E_2}\right)_{E_2=\bar{E_2}}\frac{\partial E_2}{\partial E_1}\Omega_1(E_1) = 0 \tag{2.5}$$

Remember that $\Delta E_1 = \Delta E_2$ at each time interval, therefore

$$\left(\frac{\partial \ln\Omega_1(E_1)}{\partial E_1}\right)_{E_1=\bar{E_1}} = \left(\frac{\partial \ln\Omega_2(E_2)}{\partial E_2}\right)_{E_2=\bar{E_2}} \tag{2.6}$$

Thus, our condition for equilibrium reduces to the equality of parameter β_1 and β_2:

$$\beta \equiv \left(\frac{\partial \ln\Omega(E)}{\partial E}\right)_{E=\bar{E}} \tag{2.7}$$

or a more complete version as follows,

$$\beta \equiv \left(\frac{\partial \ln\Omega(N,V,E)}{\partial E}\right)_{N,V,E=\bar{E}} \tag{2.8}$$

Therefore, we find when two systems are into thermal contact, the exchange of heat continues until the equilibrium E_1, E_2 reach some values. This happens only when the respective values of β_1 and β_2 become equal. It is then natural to expect that the parameter β is somehow related to T. To determine this relationship, we recall the thermodynamic formula

$$\left(\frac{\partial S}{\partial E}\right)_{N,V} = \frac{1}{T} \tag{2.9}$$

Comparing eq. 2.8 and 2.9, we find

$$\left(\frac{\Delta S}{\Delta \ln\Omega}\right) = \frac{1}{\beta T} = \text{const} \tag{2.10}$$

This correspondence was firstly established by Boltzmann. It was Planck who first wrote the explicit formula

$$S = k \ln\Omega \tag{2.11}$$

It means that the absolute value of the entropy of a given physical system in terms of the total number of microstates accessible to it conformity with the given macrostate, which provides a bridge between micro and macroscopic.

2.4 More complete contact

Let's continue to examine a more elaborate exchange between A_1 and A_2.

not only,

$$\left(\frac{\partial \ln\Omega_1(E_1)}{\partial E_1}\right)_{E_1=\bar{E_1}} = \left(\frac{\partial \ln\Omega_2(E_2)}{\partial E_2}\right)_{E_2=\bar{E_2}} \tag{2.12}$$

but also

$$\left(\frac{\partial \ln\Omega_1(V_1)}{\partial V_1}\right)_{V_1=\bar{V_1}} = \left(\frac{\partial \ln\Omega_2(V_2)}{\partial V_2}\right)_{V_2=\bar{V_2}} \tag{2.13}$$

Our conditions for equilibrium now take the form of an equality between the pair of (β, η)

$$\eta \equiv \left(\frac{\partial \ln\Omega(N,V,E)}{\partial V}\right)_{N,E,V=\bar{V}} \tag{2.14}$$

Similarly, there might be exchanges between particles, while need another parameter ζ,

$$\zeta \equiv \left(\frac{\partial \ln\Omega(N,V,E)}{\partial N}\right)_{V,E,N=\bar{N}} \tag{2.15}$$

To determine the physical meaning of the parameters η and ζ, we make use of the thermodynamic identity.

$$dE = TdS - PdV + \mu dN \tag{2.16}$$

so

$$\begin{aligned}\beta &= \frac{1}{kT} \\ \eta &= \frac{P}{kT} \\ \zeta &= -\frac{\mu}{kT}\end{aligned} \tag{2.17}$$

From the macroscopic view, the equilibrium is reached when

$$\begin{aligned}T_1 &= T_2 \\ P_1 &= P_2 \\ \mu_1 &= \mu_2\end{aligned} \tag{2.18}$$

This is identical to the ones following from statistical considerations. The evaluations of P, μ, T indeed requires that energy E be expressed as a function of N, V, E, this should, in principle be possible once S is known.

For instance,

$$\begin{aligned}\left(\frac{\partial S}{\partial E}\right)_{N,V} &= \frac{1}{T} \\ \left(\frac{\partial S}{\partial V}\right)_{N,V} &= \frac{P}{T} \\ \left(\frac{\partial S}{\partial N}\right)_{N,V} &= \frac{-\mu}{T}\end{aligned} \tag{2.19}$$

The rest of thermodynamic quantities follow straightforwardly.

$$\begin{aligned}F &= E - TS \\ G &= F + PV = E - TS - PV = \mu N \\ H &= E + PV = G + TS\end{aligned} \tag{2.20}$$

$$\begin{aligned}C_V &= T(\frac{\partial S}{\partial T})_{N,V} = (\frac{\partial E}{\partial T})_{N,V} \\ C_P &= T(\frac{\partial S}{\partial T})_{N,P} = (\frac{\partial H}{\partial T})_{N,P}\end{aligned} \tag{2.21}$$